

LLM Architecture Evolution

From the Initial Transformer to microGPT, GPT-1, GPT-2, and GPT-3

Prepared with ChatGPT

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Roadmap

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- 2 microGPT
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Core Idea of the Initial Transformer

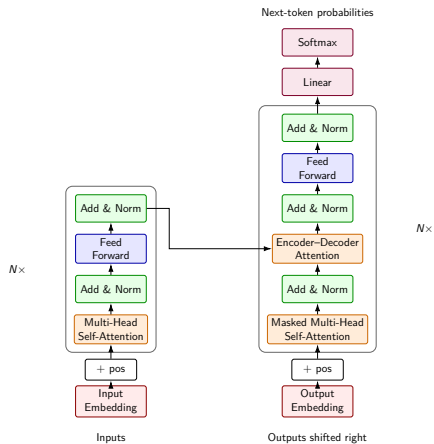
Main contribution

The paper *Attention Is All You Need* introduced the **Transformer** [Vaswani et al., 2017]: a sequence model built from

self-attention + feed-forward layers + residual connections + normalization.

- Encoder–decoder architecture for sequence-to-sequence tasks such as translation.
- No recurrence and no convolution in the main sequence-processing path.
- Positional encoding is added because attention alone does not encode order.
- Decoder uses masked self-attention plus cross-attention to the encoder outputs.

Initial Transformer Architecture



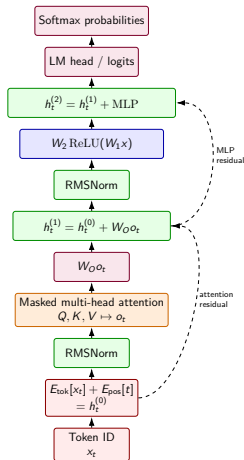
microGPT: the Minimal Educational GPT

What it is

microGPT is a tiny, dependency-free Python implementation of a decoder-only GPT. It is designed to expose the complete training and inference algorithm rather than maximize efficiency.

- Character-level language model on names.
- One Transformer layer in the example code: $L = 1$.
- Model width $d = 16$, number of heads $H = 4$, head size $d_h = 4$.
- Uses **RMSNorm** and a two-layer MLP $W_2 \text{ReLU}(W_1 x)$.
- Uses a learned token embedding E_{tok} and learned positional embedding E_{pos} .

microGPT Architecture



GPT-1: Generative Pre-Training

- Introduced in *Improving Language Understanding by Generative Pre-Training* (2018) [Radford et al., 2018].
- Decoder-only Transformer adapted from the original Transformer by discarding the encoder and cross-attention.
- Pretraining objective:

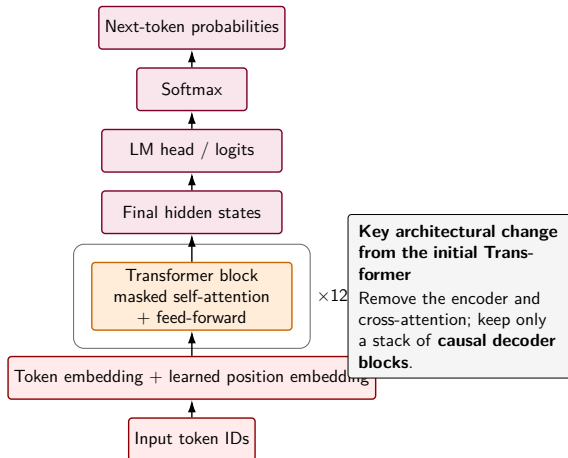
$$P(x_1, \dots, x_T) = \prod_{t=1}^T P(x_t \mid x_{<t}).$$

- Fine-tuning attaches task-specific heads after generative pretraining.

Typical GPT-1 configuration

Layers	12
Heads	12
Hidden size	768
Context	512

GPT-1 Architecture



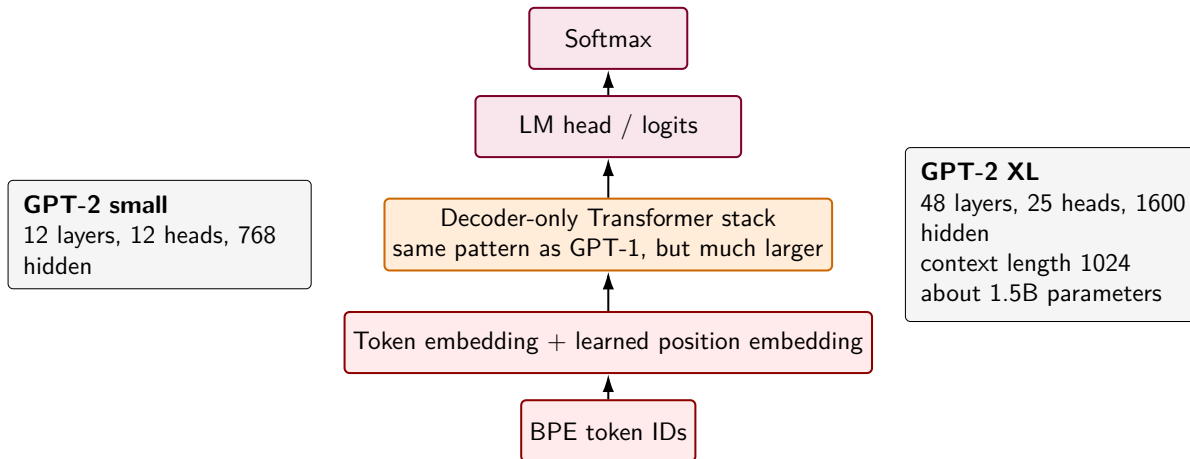
GPT-2: Scaling the Decoder-Only GPT

Main message

GPT-2 kept the same basic decoder-only architecture as GPT-1, but scaled model size, dataset size, and context handling enough to demonstrate strong zero-shot and multitask behavior [Radford et al., 2019].

- Same core pattern: token embeddings → stack of masked self-attention blocks → LM head.
- Larger context length: 1024 tokens.
- Largest published model in the paper: about 1.5B parameters.
- Emphasis shifted from “pretrain then fine-tune” toward “pretrain and prompt/use directly.”

GPT-2 as an Enlarged GPT-1



GPT-3: Scaling to Few-Shot Learning

Main message

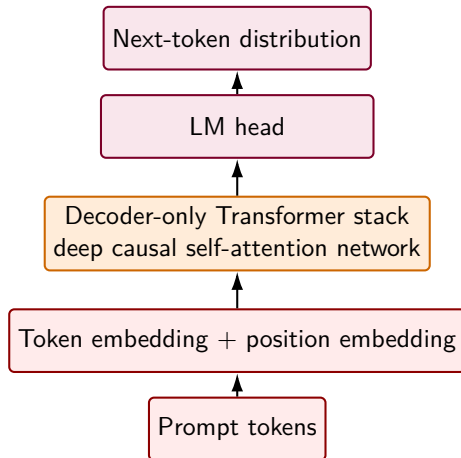
GPT-3 again retained the decoder-only Transformer pattern, but scaled it drastically [Brown et al., 2020]. The headline result was strong **in-context learning**: the model can perform new tasks from prompts alone, with zero, one, or a few demonstrations.

- Decoder-only Transformer with causal self-attention.
- Context length in the original paper: 2048 tokens.
- Largest model: 175B parameters.
- Major conceptual shift: prompting becomes a central interface to behavior.

GPT-3 Architecture: Same Skeleton, Much Larger Scale

Interpretation

Architecturally close to GPT-2, but large enough for few-shot and in-context learning to become prominent behaviors.



GPT-3 highlights

96 layers
hidden size 12,288
96 attention heads
context length 2048
175B parameters

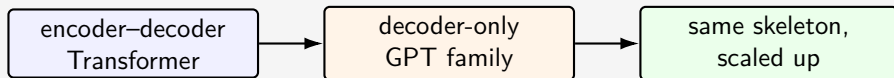
Architecture Evolution at a Glance

Model		Core architecture	Objective	Scale	Shift
Initial Trans-former		Encoder-decoder	Sequence-to-sequence	6 encoder + 6 de-coder layers	Attention replaces recur-rence
microGPT		Tiny decoder-only GPT	Character-level next-token prediction	1 layer, $d = 16$, 4 heads	Minimal educational im-plementation
GPT-1		Decoder-only Trans-former	Generative pretraining + fine-tuning	12 layers, $d = 768$	Transfer via pretraining
GPT-2		Larger decoder-only Transformer	Large-scale next-token prediction	Up to 1.5B params	Zero-shot behavior emerges
GPT-3		Much larger decoder-only Transformer	Massive next-token pre-diction	175B params	Few-shot / in-context learning

References: Vaswani et al. (2017), Radford et al. (2018, 2019), Brown et al. (2020).

The Main Structural Trend

From the initial Transformer to GPT-3



- The original Transformer introduced the attention-based architecture.
- GPT keeps only the causal decoder side.
- microGPT strips that architecture down to its essentials.
- GPT-1, GPT-2, and GPT-3 mostly differ by **scale**, training data, and behavior induced by scale.

References I



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Language Models are Few-Shot Learners.

arXiv:2005.14165, 2020.

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Andrej Karpathy.

Minimal GPT-style educational code used as the microGPT reference in these slides.

User-provided code snippet in this study session.

LLM Architecture Evolution

From the initial Transformer to GPT-3

Questions?